Specifications for the Description of Soybean Germplasm Resources

Author-prepared English reference translation

This reference translation follows the Chinese crop descriptor specification used for the CropMaestro study. It is not an official English edition issued by the source platform. Original section numbers and category-code mappings are retained.

1 Scope

These specifications define the descriptors for soybean germplasm resources and their grading criteria.

These specifications apply to the collection, curation and conservation of soybean germplasm resources; the development of data standards and data quality control specifications; and the establishment of databases and information-sharing network systems.

2 Normative references

The provisions of the following documents become provisions of these specifications through reference herein. For dated references, subsequent amendments (excluding corrections of errors) or revised editions do not apply to these specifications. However, parties entering into agreements based on these specifications are encouraged to consider whether the latest editions of these documents may be used. For undated references, the latest edition applies.

ISO 3166 Codes for the Representation of Names of Countries

GB/T 2659 Codes for the Names of Countries and Regions of the World

GB/T 2260 Codes for the Administrative Divisions of the People's Republic of China

GB/T 12404 Codes for Institutional Affiliation

GB/T1.1-2002 Directives for Standardization - Part 1: Rules for the Structure and Drafting of Standards

GB/T 3543.6-1995 Rules for Agricultural Seed Testing

GB 4404.2-1996 Seeds of Food Crops - Legumes

3 Terms and definitions

3.1 Soybean

Soybean belongs to the genus Glycine, subfamily Papilionoideae, family Leguminosae. The genus Glycine comprises two subgenera, Glycine and Soja. Subgenus Glycine comprises 16 species: Glycine albicans; G. arenaria; G. argyrea; G. canescens; G. clandestina; G. curvata; G. cyrtoloba; G. falcata; G. hirticaulis; G. lactovirens; G. latifolia; G. latrobeana; G. microphylla; G. pindanica; G. tabacina; and G. tomentella. Subgenus Soja comprises two species, annual wild soybean (G. soja Siebold & Zucc.) and cultivated soybean (G. max (L.) Merr.).

Seeds of cultivated soybean contain approximately 40% protein and 20% fat. Soybean is both an important food crop and a major oilseed crop. It is also used as livestock feed and as a raw material for food processing and light industry, and occupies an important position in the national economy.

3.2 Soybean germplasm resources

Soybean landraces, bred varieties, breeding lines, materials introduced from abroad, wild soybean resources, genetic stocks and other materials.

3.3 Basic information

Information describing the basic attributes of soybean germplasm resources, including the national unified identifier, germplasm name, scientific name, origin and germplasm type.

3.4 Morphological and biological characteristics

Characteristics of soybean germplasm resources such as phenological stages, botanical morphology and yield traits.

3.5 Quality characteristics

Crude protein content, crude fat content, amino acid content, fatty acid content, isoflavone content and other quality characteristics of soybean germplasm resources.

3.6 Abiotic stress resistance

The ability of soybean germplasm resources to adapt to or withstand various abiotic stresses, including cold, drought and salinity.

3.7 Disease and insect resistance

The ability of soybean germplasm resources to adapt to or withstand various biotic stresses, including soybean mosaic virus disease, soybean downy mildew and soybean cyst nematode disease.

4 Basic information

4.1 National unified identifier

The unique identifier of a germplasm accession. The national unified identifier for domestic cultivated soybean germplasm consists of 'ZDD' followed by a five-digit serial number; that for cultivated soybean germplasm introduced from abroad consists of 'WDD' followed by a five-digit serial number; and that for wild soybean consists of 'ZYD' followed by a five-digit serial number.

The unified identifier for vegetable soybean conserved at the Institute of Vegetables and Flowers, Chinese Academy of Agricultural Sciences, consists of 'V07F' followed by a four-digit serial number.

4.2 Genebank accession number

The accession number of soybean germplasm in the National Crop Germplasm Long-Term Genebank. The long-term genebank accession number consists of 'I2A' followed by a five-digit serial number for cultivated soybean, and 'I2B' followed by a five-digit serial number for wild soybean.

The long-term genebank accession number for vegetable soybean conserved at the Institute of Vegetables and Flowers, Chinese Academy of Agricultural Sciences, consists of 'II7F' followed by a four-digit serial number.

4.3 Introduction number

The number assigned to soybean germplasm when it is introduced from abroad.

4.4 Collecting number

The number assigned to soybean germplasm when it is collected in the wild.

4.5 Germplasm name

The Chinese name of the soybean germplasm.

4.6 Foreign-language germplasm name

The foreign-language name of germplasm introduced from abroad, or the name in Hanyu Pinyin of domestic germplasm.

4.7 Family name

Leguminosae (the legume family).

4.8 Genus name

Glycine (the soybean genus).

4.9 Scientific name

The scientific name of soybean: Glycine max (L.) Merr. for cultivated soybean, and G. soja Siebold & Zucc. for annual wild soybean.

4.10 Country of origin

The name of the country, region or international organization of origin of the soybean germplasm.

4.11 Province of origin

The province of origin of domestic soybean germplasm; for germplasm introduced from abroad, the name of the first-level administrative division of its country of origin.

4.12 Locality of origin

The names of the county, township and village of origin of domestic soybean germplasm.

4.13 Elevation

The elevation of the locality of origin of the soybean germplasm. Unit: m.

4.14 Longitude

The longitude of the locality of origin of the soybean germplasm, expressed in degrees (°) and minutes (′). The format is DDDFF, where DDD denotes degrees and FF denotes minutes. East longitudes are positive and west longitudes are negative.

4.15 Latitude

The latitude of the locality of origin of the soybean germplasm, expressed in degrees (°) and minutes (′). The format is DDFF, where DD denotes degrees and FF denotes minutes. North latitudes are positive and south latitudes are negative.

4.16 Source

For soybean germplasm introduced from abroad, the name of the source country, region or international organization; for domestic germplasm, the names of the source province and county.

4.17 Holding institution

The name of the institution that originally conserved the soybean germplasm before its submission to the National Crop Germplasm Long-Term Genebank.

4.18 Holding institution accession number

The accession number assigned to the soybean germplasm by its original holding institution.

4.19 Pedigree

The genetic relationships of a bred soybean variety or line.

4.20 Breeding institution

The name of the institution or individual that bred the soybean variety or line.

4.21 Year of development

The year in which the soybean variety or line was developed.

4.22 Breeding method

The method used to develop the soybean variety or line.

4.23 Germplasm type

Soybean germplasm types are divided into six categories.

1	Annual wild species
2	Perennial wild species
3	Landrace
4	Bred variety
5	Breeding line
6	Genetic stock
7	Other

4.24 Use type

Soybean germplasm is divided into three categories according to use.

1	Grain use
2	Vegetable use
3	Feed use

4.25 Sowing season

Soybean germplasm is divided into four categories according to sowing season.

1	Spring
2	Summer
3	Autumn
4	Winter

4.26 Ecological region

Soybean germplasm is divided into three ecological regions according to the light and temperature conditions of its source locality.

1	Northern region
2	Huang-Huai-Hai region
3	Southern region

4.27 Maturity group

Chinese soybean germplasm resources are divided into 12 maturity groups according to their responses to light and temperature.

1	000
2	00
3	0
4	I
5	II
6	III
7	IV
8	V
9	VI
10	VII
11	VIII
12	IX

4.28 Image

The filename of an image of the soybean germplasm. The image format is .jpg.

4.29 Observation site

The name of the site where the morphological and biological characteristics of the soybean germplasm are observed.

5 Morphological and biological characteristics

5.1 Flower color

The color of the petals on the day the flower opens.

1	White
2	Purple

5.2 Inflorescence length

The length of the inflorescence from its point of attachment to its tip at the beginning of pod formation.

1	Short
2	Intermediate
3	Long

5.3 Seed-coat bloom

The presence or absence of seed-coat bloom on normally mature seeds (see Figure 1).

0	Absent
1	Present

Figure 1. Seed-coat bloom



0 1

0 = absent; 1 = present. The surface coating illustrated in the Chinese source is rendered as seed-coat bloom; see the source note for 5.3.

5.4 Seed color

The seed coat color of normally mature seeds.

1	Yellow
2	Green
3	Black
4	Brown
5	Bicolored

5.5 Wild soybean seed color

The seed coat color of normally mature wild soybean seeds.

1	Yellow
2	Green
3	Black
4	Brown
5	Bicolored

5.6 Seed coat luster

The degree of luster of the seed coat of mature seeds.

0	Absent
1	Slight
2	Strong

5.7 Seed shape

The shape of mature seeds.

1	Spherical
2	Oblate
3	Ellipsoidal
4	Flattened ellipsoidal
5	Elongated ellipsoidal
6	Kidney-shaped

5.8 Seed size

Seed size is divided into five grades according to the weight of 100 mature seeds.

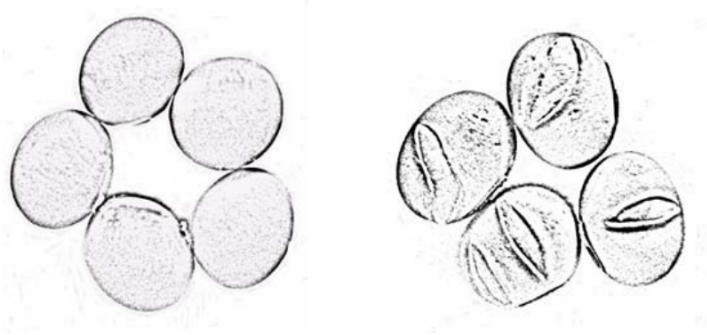
1	Very small
2	Small
3	Medium
4	Large
5	Very large

5.9 Seed-coat cracking

The degree of natural cracking of the surface of mature seeds (see Figure 2).

1	No cracking
3	Slight
5	Moderate
7	Prone to cracking

Figure 2. Seed-coat cracking



1 Seed coat intact　　2 Seed coat cracked

Figure labels 1 and 2 are retained from the Chinese source. The descriptor grades in 5.9 are 1, 3, 5 and 7.

5.10 Cotyledon color

The color of the cotyledons of mature seeds.

1	Yellow
2	Green

5.11 Hilum color

The color of the hilum of mature seeds.

1	Yellow
2	Light brown
3	Brown
4	Dark brown
5	Blue
6	Light black
7	Black

5.12 Pubescence color

The color of the pubescence on the upper-middle part of the stem and on the pod wall at maturity.

1	Gray
2	Brown

5.13 Pubescence density

The density of pubescence on the upper-middle part of the stem at maturity.

0	Absent
1	Sparse
2	Intermediate
3	Dense

5.14 Pubescence orientation

The orientation of pubescence on fully developed leaves at full flowering.

1	Erect
2	Inclined
3	Appressed

5.15 Pod color

The color of the pods at maturity.

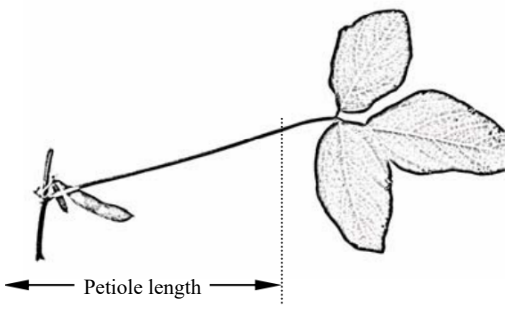
1	Gray-brown
2	Yellow-brown
3	Brown
4	Dark brown
5	Black

5.16 Petiole length

Petiole length is divided into two categories according to the length of the petiole of a fully developed leaf (see Figure 3).

1	Short
2	Long

Figure 3. Petiole length

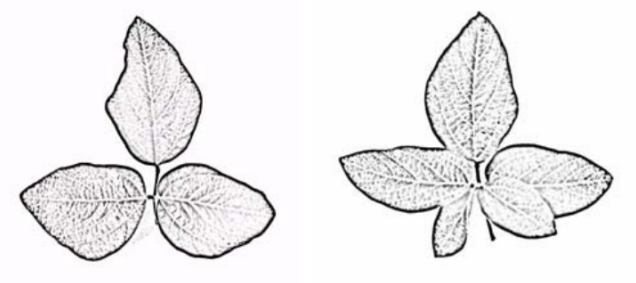


5.17 Number of leaflets

The number of leaflets per fully developed compound leaf in the upper-middle part of the plant at full flowering (see Figure 4).

1	3 leaflets
2	4-6 leaflets
3	At least 7 leaflets

Figure 4. Number of leaflets



1 2

1 = 3 leaflets; 2 = 4-6 leaflets. Only these two examples are shown in the source figure.

5.18 Leaflet shape

The shape of the terminal leaflet of a fully developed trifoliolate leaf in the upper-middle part of the plant at full flowering (see Figure 5).

1	Lanceolate
2	Ovate
4	Elliptic
4	Round

Figure 5. Leaflet shape

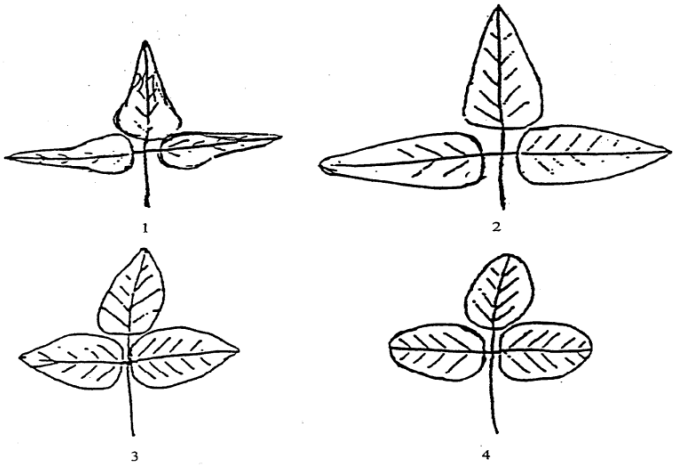


Figure labels: 1 lanceolate; 2 ovate; 3 elliptic; 4 round. These figure labels differ from the descriptor codes in 5.18, which are printed as 1, 2, 4, 4.

5.19 Leaf color

The color of leaves in the upper-middle part of the plant at full flowering.

1	Light green
2	Green
3	Dark green

5.20 Leaflet size

Leaflet size is divided into three categories according to the area of the terminal leaflet of a fully developed trifoliolate leaf in the upper-middle part of the plant at full flowering.

1	Small
2	Medium
3	Large

5.21 Leaf shedding

The degree of leaf shedding when plants are harvested at maturity.

1	No leaf shedding
2	Partial leaf shedding
3	Complete leaf shedding

5.22 Growth habit

The growth form of the main stem during the middle and late stages of plant growth and development.

1	Erect
3	Semi-erect
5	Semi-vining
7	Vining

5.23 Stem termination habit

The pod-bearing habit of plants at maturity (see Figure 6).

3	Indeterminate
5	Semideterminate
7	Determinate

Figure 6. Stem termination habit

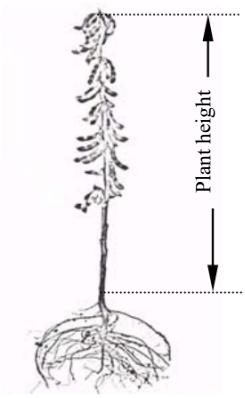


3 = indeterminate; 5 = semideterminate; 7 = determinate.

5.24 Plant height

The height from the cotyledonary node to the growing point of the plant at maturity. Unit: cm (see Figure 7).

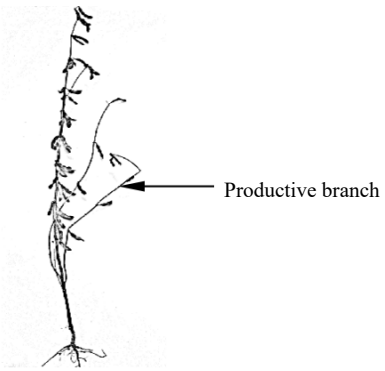
Figure 7. Plant height



5.25 Number of productive branches

The number of primary branches with at least two stem nodes and at least one mature pod on a plant at maturity. Unit: branches (see Figure 8).

Figure 8. Number of productive branches

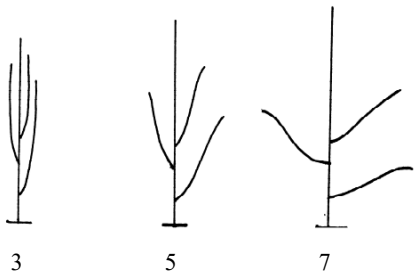


5.26 Plant architecture

The degree of compactness of the lower branches relative to the main stem at maturity (see Figure 9).

3	Compact
5	Semi-spreading
7	Spreading

Figure 9. Plant architecture



3 = compact; 5 = semi-spreading; 7 = spreading.

5.27 Hypocotyl color

The color of the hypocotyl when the first trifoliolate leaf has unfolded.

1	Green
2	Purple

5.28 Main stem prominence

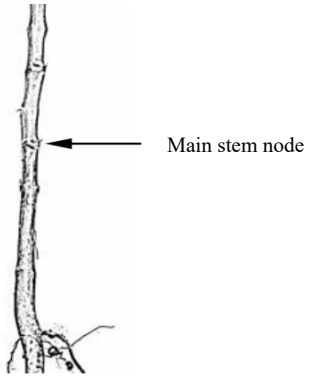
The prominence of the main stem relative to the branches in wild soybean at full flowering.

1	Prominent
2	Not prominent

5.29 Number of main stem nodes

The number of nodes from the cotyledonary node to the tip of the main stem at plant maturity. Unit: nodes (see Figure 10).

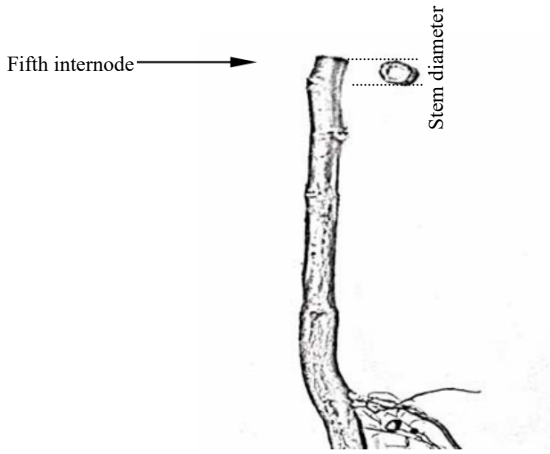
Figure 10. Main stem nodes



5.30 Stem diameter

The diameter of the fifth internode of the main stem at plant maturity. Unit: mm (see Figure 11).

Figure 11. Stem diameter

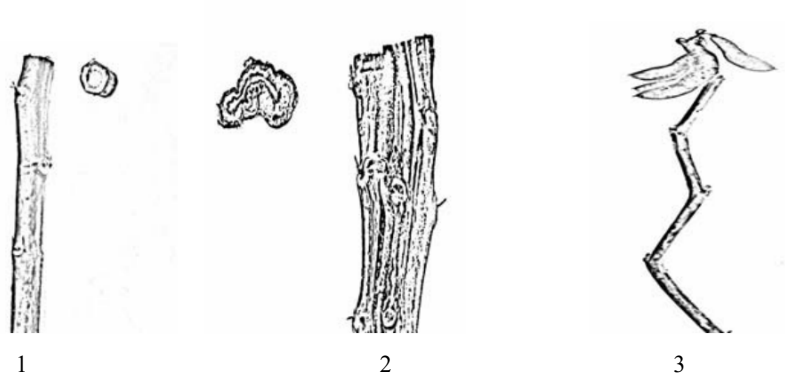


5.31 Stem shape

The shape of the main stem at plant maturity (see Figure 12).

1	Normal stem
2	Flattened stem
3	Bent stem

Figure 12. Main stem shape



1 = normal stem; 2 = flattened stem; 3 = bent stem.

5.32 Stem strength

The toughness of the stem during the late stages of plant growth and development.

1	Weak
2	Intermediate
3	Strong

5.33 Lodging

The degree of plant inclination during the late stages of plant growth and development.

1	None
3	Slight
5	Moderate
7	Heavy
9	Severe

5.34 Root nodules

The presence or absence of root nodules at the beginning of flowering (see Figure 13).

0	Absent
1	Present

Figure 13. Root nodules



0 1

0 = absent; 1 = present.

5.35 Number of pods per plant

The actual number of seed-bearing pods per plant at maturity. Unit: pods.

5.36 Number of pods per 500 g

The number of fresh pods in a 500 g sample of pods at fresh-pod harvest of vegetable soybean. Unit: pods.

5.37 Pod size

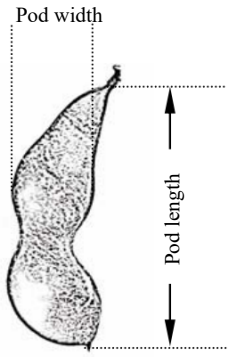
Pod size is divided into three categories according to pod length at maturity.

1	Small
2	Medium
3	Large

5.38 Pod length

The length of a pod at fresh-pod harvest of vegetable soybean. Unit: cm (see Figure 14).

Figure 14. Pod length and width



5.39 Pod width

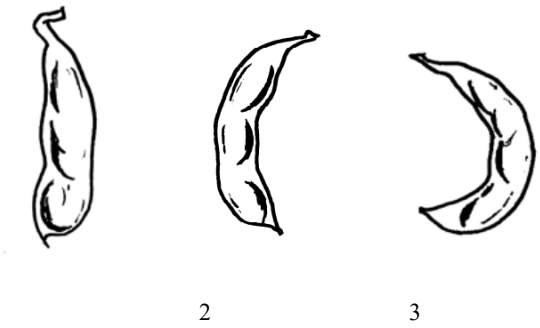
The width at the widest part of a pod at fresh-pod harvest of vegetable soybean. Unit: cm (see Figure 14).

5.40 Pod shape

The shape of pods on the upper-middle part of the main stem from peak seed filling to maturity (see Figure 15).

1	Straight
2	Sickle-shaped
3	Bow-shaped

Figure 15. Pod shape



1 = straight; 2 = sickle-shaped; 3 = bow-shaped.

5.41 Height of the lowest pod

The height from the cotyledonary node to the lowest pod, the pod-bearing inflorescence or the point of attachment of the first productive branch at maturity. Unit: cm (see Figure 16).

Figure 16. Height of the lowest pod



5.42 Pod shattering

The extent and grade of natural pod dehiscence at maturity.

3	No shattering
5	Slight shattering
7	Moderate shattering
9	Prone to shattering

5.43 Number of seeds per plant

The actual number of seeds per plant at maturity, excluding unfilled seeds but including immature, diseased and insect-damaged seeds. Unit: seeds.

5.44 Number of seeds per pod

The average number of seeds per pod for an individual plant at maturity. Unit: seeds.

5.45 Seed weight per plant

The weight of seeds actually harvested from an individual plant at maturity. Unit: g.

5.46 Fresh 100-seed weight

The weight of 100 intact seeds at fresh-pod harvest of vegetable soybean. Unit: g.

5.47 100-seed weight

The weight of 100 intact seeds after maturity. Unit: g.

5.48 Seed yield per unit area

The net weight of seeds harvested per unit area at maturity. Unit: kg/hm^2.

5.49 Calendar months spanned by the growing period

The calendar months encompassed by the growing period of the soybean germplasm.

5.50 Days to maturity

For spring soybean, the number of days from the day after emergence through the date of maturity; for summer and autumn soybean, the number of days from the day after sowing through the date of maturity. Unit: d.

5.51 Sowing date

The date on which seeds are sown for the characterization of morphological and biological characteristics of soybean germplasm resources. Expressed as year, month and day in the format YYYYMMDD.

5.52 Emergence date

The date when the cotyledons of 50% of plants in the plot have emerged above the soil. Expressed as year, month and day in the format YYYYMMDD.

5.53 Flowering date

The date when 50% of plants in the plot are flowering. Expressed as year, month and day in the format YYYYMMDD.

5.54 Pod formation date

The date when 50% of plants in the plot have formed young pods at least 2 cm long. Expressed as year, month and day in the format YYYYMMDD.

5.55 Seed-filling date

The date when 50% of plants in the plot have pods that, when laid flat, show clearly protruding seeds. Expressed as year, month and day in the format YYYYMMDD.

5.56 Maturity date

The date when 50% of plants have 95% of their pods in the mature color and begin to rattle when shaken. Expressed as year, month and day in the format YYYYMMDD.

6 Quality characteristics

6.1 Crude protein content

The total protein content of seeds. Expressed as %.

6.2 Lysine content

The lysine content of seeds. Expressed as %.

6.3 Tryptophan content

The tryptophan content of seeds. Expressed as %.

6.4 Phenylalanine content

The phenylalanine content of seeds. Expressed as %.

6.5 Methionine content

The methionine content of seeds. Expressed as %.

6.6 Threonine content

The threonine content of seeds. Expressed as %.

6.7 Isoleucine content

The isoleucine content of seeds. Expressed as %.

6.8 Leucine content

The leucine content of seeds. Expressed as %.

6.9 Valine content

The valine content of seeds. Expressed as %.

6.10 Cystine content

The cystine content of seeds. Expressed as %.

6.11 11S/7S ratio

The ratio of the 11S globulin content to the 7S globulin content of seeds.

6.12 28K allergenic protein

The deficiency status of the 28K allergenic protein subunit in seeds.

0	Absent
1	Present

6.13 30K allergenic protein

The deficiency status of the 30K allergenic protein subunit in seeds.

0	Absent
1	Present

6.14 Kunitz trypsin inhibitor

The deficiency status of the Kunitz-type trypsin inhibitor in seeds.

0	Absent
1	Present

6.15 Crude fat content

The fat content of seeds. Expressed as %.

6.16 Stearic acid content

The content of the saturated fatty acid stearic acid (18:0) in seeds. Expressed as %.

6.17 Palmitic acid content

The content of the saturated fatty acid palmitic acid (16:0) in seeds. Expressed as %.

6.18 Oleic acid content

The oleic acid (18:1) content of seeds. Expressed as %.

6.19 Linoleic acid content

The linoleic acid (18:2) content of seeds. Expressed as %.

6.20 Linolenic acid content

The linolenic acid (18:3) content of seeds. Expressed as %.

6.21 Lipoxygenase

The deficiency status of lipoxygenases Lox1, Lox2 and Lox3 in seeds.

1	Lox1 absent
2	Lox2 absent
3	Lox3 absent
4	Lox1 and Lox2 absent
5	Lox1 and Lox3 absent
6	Lox2 and Lox3 absent
7	Lox1, Lox2 and Lox3 absent
8	No Lox deficiency

6.22 Sugar content

The sugar content of seeds. Expressed as %.

6.23 Isoflavone content

The isoflavone content of seeds. Unit: 10^{-2} mg/g.

7 Abiotic stress resistance

7.1 Salt tolerance at germination

The ability of soybean to tolerate salt stress at the germination stage.

1	Tolerant
2	Fairly tolerant
3	Moderately tolerant
4	Relatively sensitive
5	Sensitive

7.2 Salt tolerance at the seedling stage

The ability of soybean to tolerate salt stress, mainly from sodium chloride, at the seedling stage.

0	Highly tolerant
1	Tolerant
2	Fairly tolerant
3	Moderately tolerant
4	Relatively sensitive
5	Sensitive

7.3 Salt tolerance at flowering and podding

The ability of soybean to tolerate salt stress, mainly from sodium chloride, during flowering and podding.

1	Tolerant
2	Fairly tolerant
3	Moderately tolerant
4	Relatively sensitive
5	Sensitive

7.4 Drought tolerance at germination

The ability of soybean to tolerate drought stress at the germination stage.

1	Tolerant
2	Fairly tolerant
3	Moderately tolerant
4	Relatively sensitive
5	Sensitive

7.5 Drought tolerance of mature plants

The ability of soybean to tolerate drought stress under natural drought conditions.

1	Tolerant
2	Fairly tolerant
3	Moderately tolerant
4	Relatively sensitive
5	Sensitive

7.6 Cold tolerance at germination

The ability of soybean to tolerate chilling stress during germination.

1	Tolerant
2	Fairly tolerant
3	Relatively sensitive
4	Sensitive

7.7 Aluminum tolerance in acidic soil

The ability of soybean plants to tolerate aluminum ion toxicity in acidic soil.

1	Tolerant
2	Sensitive

7.8 Acid rain tolerance

The ability of soybean plants to tolerate acid rain stress.

1	Tolerant
2	Fairly tolerant
3	Moderately tolerant
4	Relatively sensitive
5	Sensitive

8 Disease and insect resistance

8.1 Soybean rust resistance

The degree of resistance of soybean plants to the soybean rust pathogen (Phakopsora pachyrhizi Sydow).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.2 Frogeye leaf spot resistance

The degree of resistance of soybean plants to the frogeye leaf spot pathogen (Cercospora sojina Hara).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.3 Soybean downy mildew resistance

The degree of resistance of soybean plants to the downy mildew pathogen (Peronospora manshurica (Naumov) Syd.).

0	Immune (I)
1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.4 Purple seed stain resistance

The degree of resistance of soybean seeds to the purple seed stain pathogen (Cercospora kikuchii (Matsumoto & Tomoyasu) Gardner).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.5 Bacterial blight resistance

The degree of resistance of soybean leaves to the bacterial blight pathogen (Pseudomonas syringae pv. glycinea).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.6 Soybean mosaic virus resistance

The degree of resistance of soybean leaves to soybean mosaic virus (SMV).

0	Immune (I)
1	Highly resistant (HR)
3	Resistant (R)
7	Susceptible (S)
9	Highly susceptible (HS)

8.7 Phytophthora root rot resistance

The degree of resistance of soybean plants to Phytophthora sojae Kaufm. & Gerd.

3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)

8.8 Soybean cyst nematode resistance

The degree of resistance of soybean plants to disease caused by the soybean cyst nematode (Heterodera glycines Ichinohe).

0	Immune (I)
1	Highly resistant (HR)
3	Resistant (R)
7	Susceptible (S)
9	Highly susceptible (HS)

8.9 Soybean pod borer resistance

The degree of resistance of soybean seeds to the soybean pod borer (Leguminivora glycinivorella (Matsumura)).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.10 Soybean aphid resistance

The degree of resistance of soybean plants to the soybean aphid (Aphis glycines Matsumura).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.11 Legume pod borer resistance

The degree of resistance of soybean plants to the legume pod borer (Etiella zinckenella (Treitschke)).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.12 Soybean stem fly resistance

The degree of resistance of soybean plants to the soybean stem fly (Melanagromyza sojae (Zehntner)).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.13 Resistance to leaf-feeding insects

The degree of resistance of soybean leaves to leaf-feeding insects, also termed insect defoliators or foliar-feeding insects.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

9 Other characteristics

9.1 Fingerprint identifier

The identifier of a soybean germplasm accession in a fingerprint profile.

9.2 Core collection accession number

The accession number of soybean germplasm in the core collection.

9.3 Remarks

Specific explanations of special descriptors or special codes for the soybean germplasm.

Translator's notes on the source

2: The reference title for GB/T 3543.6-1995 is reproduced at the level of detail supplied in the Chinese source.

3.1; 4.9; 8: The historical classification and species count are retained. Verified spelling and author corrections are applied to Glycine lactovirens; G. soja Siebold & Zucc.; G. max (L.) Merr.; Peronospora manshurica (Naumov) Syd.; Cercospora kikuchii (Matsumoto & Tomoyasu) Gardner; Phytophthora sojae Kaufm. & Gerd.; Leguminivora glycinivorella (Matsumura); Etiella zinckenella; and Melanagromyza sojae (Zehntner). GBIF and EPPO records support these corrections. Historical taxa have not been replaced with modern accepted names.

4.2: The vegetable-soybean long-term genebank prefix 'II7F' is reproduced as printed in the Chinese source.

4.23: The Chinese source states six categories but lists seven codes (1-7). Both the stated count and all seven codes are retained.

5.3; Figure 1: Seed-coat bloom is the English rendering used here for the surface coating illustrated in Figure 1. USDA GRIN confirms bloom as an English soybean descriptor for a powdery coating adhering to the seed coat. The correspondence with the Chinese term is inferred from context; an official bilingual equivalence has not been established. The source does not specify the coating composition.

5.9; Figure 2: Figure 2 labels the two examples 1 and 2, whereas 5.9 uses grades 1, 3, 5 and 7. The source figure labels and descriptor grades are both retained.

5.18; Figure 5: The descriptor codes in 5.18 are printed as 1, 2, 4, 4, while Figure 5 uses 1, 2, 3, 4. This discrepancy is retained; the distinct labels elliptic and round are preserved.

5.41: The source gives three alternative reference points for the lowest-pod height but does not specify how to choose between them. All three are retained.

5.49: The source does not specify whether this descriptor records the calendar months spanned or their number. No numerical data type is inferred.

5.55: The phrase 'when laid flat' follows the source wording, whose observation procedure is not further explained. It is not treated as a separate flattened-pod character. The 50% threshold refers to plants.

6.12: The source describes deficiency of the 28K allergenic protein but labels the states only 'absent' (0) and 'present' (1). It does not specify whether these labels refer to the protein or to its deficiency. The code polarity therefore remains unresolved.

6.13: The source describes deficiency of the 30K allergenic protein but labels the states only 'absent' (0) and 'present' (1). It does not specify whether these labels refer to the protein or to its deficiency. The code polarity therefore remains unresolved.

6.14: The source describes deficiency of the Kunitz trypsin inhibitor but labels the states only 'absent' (0) and 'present' (1). It does not specify whether these labels refer to the protein or to its deficiency. The code polarity therefore remains unresolved.

7.1-7.8: Tolerance scales follow each source subsection: 7.2 includes code 0, 7.6 has four grades and 7.7 has two grades. No common scale has been imposed.

8.3; 8.6-8.8: Disease resistance scales are retained exactly: 8.3 includes 0, 1, 3, 5, 7 and 9; 8.6 and 8.8 omit code 5; and 8.7 contains only 3, 5 and 7.